Apparent Complexity of Quantized Diffusion: Contour Maps, Aliasing and Moiré

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Code and figures: https://github.com/MiguelPerezI/heat-parity-ca

Abstract

Colour the cells of a heat-equation simulation by the parity of $\lfloor q\,u \rfloor$, and the frames look like a cellular automaton with a rich transient: noise, then a lattice of concentric islands, then nothing. They are a contour map of the temperature, the rise in complexity is aliasing, and the islands are replicated centres of an undersampled Fresnel zone plate. Three elementary results make this exact. The total variation decreases. The phase-averaged perimeter of the pattern equals $\sum_e \operatorname{dist}(q|\Delta_e u|, 2\mathbb{Z})$, a triangle wave of the jump across each grid edge. And the block entropy has two limits that do not commute. The perimeter identity holds for any field. The gap between the scaled total variation $q\,\mathrm{TV}_h$ and the expected perimeter measures the aliasing exactly. Applied to the Turing spots of a glycolysis model, it shows the converse effect: aliasing hides a real growth of structure behind proxies that saturate, and it leaves a permanent static. In the faithful stage, counting contours measures the Turing eigenvalue. Complexity measures of a thresholded field report the sampling as much as the dynamics, a point with consequences for proposed laws of increasing functional information.

Keywords: aliasing; moiré; contour maps; heat equation; Turing patterns; total variation; complexity measures
MSC 2020: 35K05, 35K57, 65M06, 68U10, 00A66

1 The observation

The two-dimensional heat equation $u_t = \kappa \Delta u$ was integrated on a 64×64 grid with an explicit scheme, starting from a hot square ($u = 100$ on a 24×24 block, 0 elsewhere), and displayed as a height field in which each facet is coloured white or black according to the parity of $\lfloor 10\,u \rfloor$. The frames (Figure 1) begin uniformly black, erupt into a fine noise-like texture around the square, organize into a square-symmetric lattice of concentric islands, and finally fade to a few rings and then to a single colour. The sequence looks like a cellular automaton with a rich transient, and the islands look like eddies. Neither is the case, and the point of this note is to say precisely what they are.

Throughout, $\Omega = (0, L)^n$, u solves $u_t = \kappa \Delta u$ with homogeneous Neumann data, $q > 0$ is the quantization frequency (quantum $1/q$), $\theta \in [0, 1)$ is a phase, and the *pattern* is

$$b_\theta(x, t) = \lfloor q\,u(x, t) + \theta \rfloor \bmod 2. \tag{1}$$

The grid version samples u at the N^n nodes of the explicit scheme with reflected ghost values (the ternary-cube rule of [1] with zero flux), under $\sum_m r_m \le \frac{1}{2}$. We write E for the set of grid edges and $\Delta_e u$ for the jump of u across the edge e.

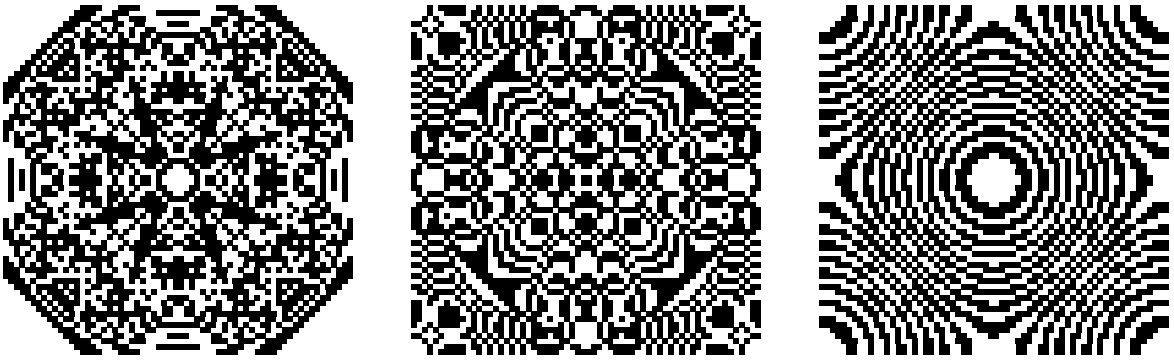


Figure 1: Parity of $\lfloor 10u \rfloor$ on the 64×64 grid at steps 150, 1000 and 2000 of the explicit scheme with $\Delta t = 2 \cdot 10^{-4}$, $\kappa = 0.2$, reflecting walls. The field has a single maximum throughout.

So (1) is the contour map of u with isotherms at the levels $(k-\theta)/q$ and alternating fill. The pattern is an *observable* of u, not a state: two fields with the same pattern evolve to different patterns, so no local rule on black and white cells reproduces the frames, and by the usual definition (a map on configurations over a finite alphabet given by a local rule [7]) this is not a cellular automaton.

2 What is monotone in the continuum

Theorem 2.1. *Let u solve the Neumann heat equation on the box Ω.*

1. *The band count $B_q(t) = \#\{k : (k-\theta)/q \in [\min u(\cdot,t), \max u(\cdot,t)]\}$ is nonincreasing for every θ.*
2. *The total variation $\mathrm{TV}(t) = \int_\Omega |\nabla u(x,t)|\, dx$ is nonincreasing.*
3. *The phase-averaged total isotherm length equals the total variation exactly:*

$$\int_0^1 \sum_{k\in\mathbb{Z}} \mathcal{H}^{n-1}\big(\{u(\cdot,t) = (k-\theta)/q\}\big)\, d\theta \;=\; q\ \mathrm{TV}(t), \tag{2}$$

and is therefore nonincreasing.

Proof. (1) is the maximum principle [5]: $\max u$ is nonincreasing and $\min u$ nondecreasing. (2) Reflect u evenly across every face of the box; the result $\tilde{u}$ is a $2L$-periodic solution of the heat equation on the torus $\mathbb{T} = (\mathbb{R}/2L\mathbb{Z})^n$, smooth for $t > 0$. Each component $\partial_m \tilde{u}$ is again a periodic heat solution, so $\nabla\tilde{u}(t) = K_{t-s} * \nabla\tilde{u}(s)$ for $t > s$, where K is the periodic heat kernel, $K \geq 0$, $\int K = 1$. Convexity of the Euclidean norm gives $|\nabla\tilde{u}(t)| \leq K_{t-s} * |\nabla\tilde{u}(s)|$ pointwise; integrating over $\mathbb{T}$ and dividing by 2^n gives $\mathrm{TV}(t) \leq \mathrm{TV}(s)$. (3) By the coarea formula, $\int_\Omega |\nabla u| = \int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, ds$; the left side of (2) is $\int_{\mathbb{R}} \mathcal{H}^{n-1}(\{u = s\})\, q\, ds$ after the substitution $s = (k-\theta)/q$, $\theta \in [0,1)$, $k \in \mathbb{Z}$. □

For a fixed phase the sum in (2) is a Riemann sum of the coarea integral, hence within $o(1)$ (as $q \to \infty$) of a nonincreasing function. In the continuum the quantized contour map has no rise of complexity, for any quantum, in the phase-averaged sense.

3 The exact perimeter formula on the grid

Let P_h be the number of grid edges whose two cells differ in colour, and $C_h = \sum_{e\in E} |\lfloor qu_a + \theta \rfloor - \lfloor qu_b + \theta \rfloor|$ the number of isotherm crossings, where a, b are the ends of e. Let $\mathrm{TV}_h = \sum_{e\in E} |\Delta_e u|$ and let $\varphi(x) = \mathrm{dist}(x, 2\mathbb{Z})$ be the triangle wave of period 2 with $\varphi(x) = x$ on $[0,1]$.

Theorem 3.1. *1. (Discrete TVD.) For each axis m, $\sum_{\mathbf{i}} |u_{\mathbf{i}+\mathbf{e}_m} - u_{\mathbf{i}}|$ is nonincreasing under the explicit scheme with reflected ghosts and $\sum_m r_m \le \frac{1}{2}$. Hence TV_h is nonincreasing.*

2. (Random phase.) With θ uniform on $[0,1)$,

$$\mathbb{E}_\theta C_h = q\ \mathrm{TV}_h, \qquad \mathbb{E}_\theta P_h = \sum_{e\in E} \varphi\big(q|\Delta_e u|\big), \tag{3}$$

both exactly.

3. Consequently: if $q|\Delta_e u| \le 1$ on every edge then $\mathbb{E}_\theta P_h = q\,\mathrm{TV}_h$ and is nonincreasing; always $\mathbb{E}_\theta P_h \le \min(q\,\mathrm{TV}_h, |E|)$; every increase of $\mathbb{E}_\theta P_h$ in time is produced by edges whose jump exceeds one quantum and is decreasing through a folded branch of φ; and $\mathbb{E}_\theta P_h$ is nonincreasing from the first time t^* at which $\max_e q|\Delta_e u| \le 1$.

Proof. (1) On the reflected periodic grid the scheme is a convex combination of translates, $u^{l+1} = \sum_j c_j\, \sigma_j u^l$ with $c_j \ge 0$, $\sum c_j = 1$, by the CFL condition. The axis difference $D_m u = \sigma_{\mathbf{e}_m} u - u$ satisfies the same recursion, and a convex combination of translates is an ℓ^1-contraction (the linear case of Harten's TVD criterion [6]), so $\|D_m u^{l+1}\|_1 \le \|D_m u^l\|_1$. Restricting the periodic sum to the box changes it by the factor 2^n. (2) Under a uniform shift the number of levels $(k-\theta)/q$ inside an interval of length ℓ is $\lfloor q\ell \rfloor$ with probability $1-\{q\ell\}$ and $\lfloor q\ell\rfloor + 1$ with probability $\{q\ell\}$. Its expectation is $q\ell$, which gives the first identity. The two ends of the edge differ in colour iff that number is odd; if $\lfloor q\ell \rfloor$ is even this has probability $\{q\ell\}$, otherwise $1-\{q\ell\}$, and both equal $\varphi(q\ell)$. (3) follows from $\varphi(x) = x$ on $[0,1]$, $\varphi \le \min(x,1)$, and the fact that φ is increasing only on the branches $[2j-1, 2j]$, $j \ge 1$, which require $x > 1$. □

Part (2) uses nothing about the dynamics: it is an identity for any real field on the grid, and only part (1) is specific to the heat equation. It therefore splits the perimeter of any quantized field into what the field does and what the sampling does.

Corollary 3.2 (Aliasing deficit)**.** *For any field u on the grid let $D_h = q\,\mathrm{TV}_h - \mathbb{E}_\theta P_h$. Then*

$$D_h = \sum_{e\in E} \psi\big(q|\Delta_e u|\big), \qquad \psi(x) = x - \varphi(x), \tag{4}$$

where ψ is nondecreasing, $\psi = 0$ on $[0,1]$ and $x - 1 \le \psi(x) \le x$. Hence $D_h \ge 0$, with $D_h = 0$ if and only if $\max_e q|\Delta_e u| \le 1$; and $\mathbb{E}_\theta P_h \le |E|$ whatever the size of TV_h, so $D_h \ge q\,\mathrm{TV}_h - |E|$.

Proof. Subtract the two identities (3). $\psi' = 1 - \varphi' \in \{0, 2\}$ almost everywhere, so ψ is nondecreasing; it vanishes where $\varphi(x) = x$, that is on $[0,1]$; and $0 \le \varphi \le 1$. □

So $\mathbb{E}_\theta P_h$ is a faithful measure of $q\,\mathrm{TV}_h$ exactly as long as no edge carries more than one quantum. Beyond that, the perimeter can rise while TV_h falls, which is the heat-equation case of this section, or stay flat while TV_h grows without bound, which is the case of Section 6.

Theorem 3.1 explains the frames. Initially the jumps across the edge of the hot square are a thousand quanta, φ is folded a thousand times, and the expected perimeter per such edge is about $\frac{1}{2}$. As the front spreads the number of edges in the folded regime grows, so the perimeter grows, although $q\,\mathrm{TV}_h$ is falling all along. When the largest jump passes below one quantum, the perimeter becomes $q\,\mathrm{TV}_h$ itself and decays with it. Table 1 shows the identity (3) against a Monte-Carlo average over 200 phases on the setup of Figure 1, and the peak at the step where the largest jump crosses one quantum.

step	$q\,\mathrm{TV}_h$	$P_h(\theta=0)$	$\overline{P_h}$ (MC)	$\sum_e \varphi(q\lvert\Delta_e u\rvert)$	$\max_e q\lvert\Delta_e u\rvert$
20	95999.7	1136	1256.2	1257.3	155.95
150	87103.5	3600	3613.3	3610.7	51.75
400	59437.5	4032	4004.1	4004.7	22.19
1000	22542.9	4232	4285.2	4286.4	5.75
1500	10006.7	3560	3552.3	3555.3	2.22
2000	4435.4	4432	4436.8	4435.4	0.92
2500	1964.6	1976	1963.5	1964.6	0.39
3000	870.0	896	867.3	870.0	0.17
4000	170.5	184	172.9	170.5	0.03

Table 1: The random-phase identity (3) on the 64×64 run of Figure 1, $q = 10$. The discrete total variation never increased; the perimeter peaks where the largest jump crosses one quantum; from then on $\overline{P_h} = q\,\mathrm{TV}_h$ to rounding.

4 Entropy: the limits do not commute

Let $H_{q,h}(t)$ be the Shannon entropy of the empirical distribution of 2×2 blocks of b_θ on the grid of spacing h.

Theorem 4.1. *Fix $t > 0$.*

1. *For fixed q, $\lim_{h\to 0} H_{q,h}(t) = 0$.*
2. *For fixed h, if $1, u(x_1,t), \dots, u(x_M,t)$ are linearly independent over $\mathbb{Q}$, where x_j are the grid nodes, then $(q\,u(x_j,t) \bmod 2)_j$ is equidistributed on $(\mathbb{R}/2\mathbb{Z})^M$ as $q \to \infty$, and the Cesàro limit over q of $H_{q,h}(t)$ is the maximum $\log_2 16 = 4$ bits.*

Proof. (1) For $t > 0$ the isotherms are smooth curves of finite total length $\Lambda_q(t)$; a 2×2 block is non-uniform only if an isotherm passes within distance $2h$ of it, so the fraction of non-uniform blocks is $O(h\Lambda_q(t))$, and the entropy of a distribution concentrated on one symbol with mass $1 - O(h)$ is $O(h \log(1/h))$. (2) is Weyl's equidistribution theorem (Kronecker–Weyl) for the sequence $q \mapsto (qu(x_j,t))_j$ on the torus, under the stated rational independence; the uniform distribution on $\{0,1\}^4$ maximizes the block entropy. □

The square-symmetric initial data of the experiment violate the independence in (2), since $u(x_j) = u(\sigma x_j)$ for the symmetries σ of the square, and the achievable entropy is reduced by the orbit structure. That is why the aliased frames look like symmetric designs rather than noise. Measured on the frames of Figure 1, the block entropy reaches 3.99 bits at step 400; this is regime (2), a statement about the sampling, not about the heat equation.

5 The islands are moiré

Where u is locally linear, $\lfloor q(a \cdot x) + \theta \rfloor \bmod 2$ on the lattice is a two-colour digital plane, whose one-dimensional sections are Sturmian words; these are the striped regions. Where u is locally quadratic, near the maximum and in the tails, the pattern is a sampled Fresnel zone plate. An undersampled zone plate replicates its centre [4, 9]: neighbouring cells agree in colour wherever $q\,h\,\partial_m u$ is near an even integer, so fake centres appear along the curves $\{qh\,\partial_m u \in 2\mathbb{Z}\}$, $m = 1, 2$. Figure 2 shows the effect in its purest form: the state at $t = 0.03$, when the field still carries 832 isotherms and the gradient at the edge of the square is about forty quanta per cell even on a 256^2 grid. The true contour map would be a set of tightly

nested rounded squares; what the lattice sees instead is a central zone plate surrounded by a ring of replica centres, each a small zone plate of its own, arranged on the curves where $qh\,\partial_m u$ passes through even integers, with the quasi-periodic stripes of digital planes between them. Every feature of this picture is a property of the sampling: the field has one maximum, and its isotherms never cross. Figure 3 shows the later state $t = 0.4$ at 64 and 256 cells with the same quantum: the island lattice is present at 64 and absent at 256, where the frame is an ordinary contour map. Table 2 gives the measures. With the quantum scaled as $1/N$ instead, the measures are constant in N (perimeter per cell 0.879, 0.894, 0.893): the texture depends on the single ratio $qh|\nabla u|$ and is scale-free undersampling.

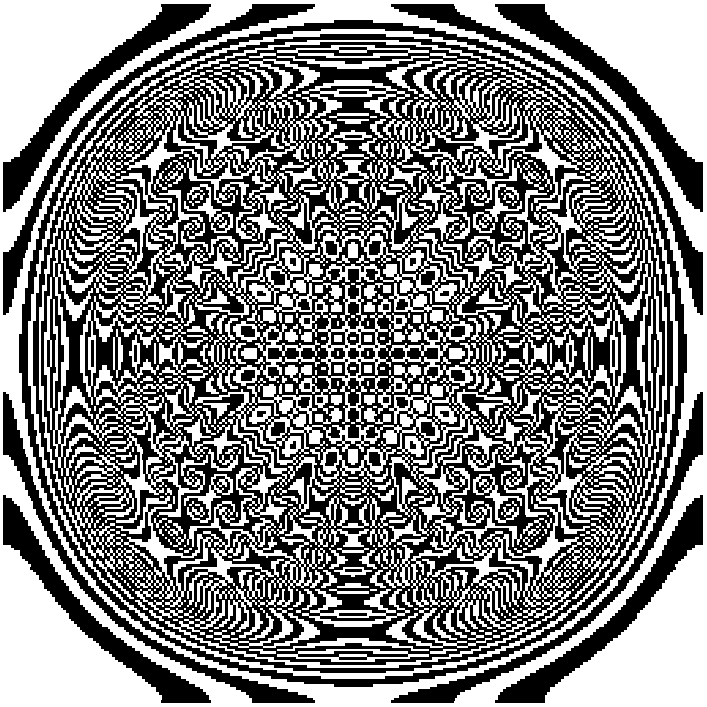


Figure 2: The parity pattern at $t = 0.03$ on a 256^2 grid, $q = 10$, 832 isotherms. An undersampled Fresnel zone plate: the central rings are the real contour map of the peak, the surrounding lattice of small concentric centres is moiré, replicas created where the local fringe frequency matches a multiple of the sampling frequency, and the striped sectors between them are digital planes. The figure is vector graphics; zooming in shows the individual cells.

N	bands	perimeter/N^2	$H_{2\times2}$ (bits)	components
64	34	1.082	3.93	905
128	36	0.564	3.45	88
256	37	0.287	2.61	92

Table 2: Resolution test at $t = 0.4$, $q = 10$ fixed. Perimeter per cell halves with each doubling, as a fixed set of smooth curves does; the islands collapse into the real rings.

6 A growing field: Turing spots in glycolysis

The heat equation is the case where every intrinsic quantity decays. For the opposite case, take a field whose total variation grows: the Turing instability [14] of the two-variable Sel'kov



Figure 3: Left: 64×64 grid at $t = 0.4$ ($q = 10$). Right: the same state on a 256×256 grid with the same quantum. The islands of the left frame are replicated zone-plate centres; the right frame is the contour map of the field, with its mirror images in the four corners produced by the reflecting walls.

model of glycolysis in the form of Ashkenazi and Othmer [15], studied by Cisneros-Ake et al. [13],

$$u_t = D_u \Delta u - u + v(a + u^2), \qquad v_t = D_v \Delta v + b - v(a + u^2), \tag{5}$$

on $[0, 1]^2$ with Neumann data. We use the parameters of their spot pattern ([13, Fig. 4]): $a = 0.1$, $b = 0.85$, $D_u = 0.0025$, $D_v = 8D_u$, 101^2 nodes ($h = 0.01$), $\Delta t = 10^{-3}$, initial data the equilibrium $(u^*, v^*) = (b,\, b/(a + b^2))$ plus 0.01 times uniform noise. On this grid exactly two modes are unstable, $\cos 3\pi x \cos 2\pi y$ and $\cos 2\pi x \cos 3\pi y$, with growth rate $\lambda = 0.01103$, in agreement with the leading eigenvalue $\lambda = 0.011$ of [13, Fig. 6]. We display u by the parity of $\lfloor qu \rfloor$ with $q = 10^4$. The quantum is chosen as an observer would choose it from the first frames, whose total range is below 10^{-3}, and it reproduces the screen captures that motivated this section. The general statement is the following.

Proposition 6.1. *Let $u(t)$ be any family of fields on the grid.*

1. *(Exponential growth.) If $u(t) = \bar{u} + c\, e^{\lambda t} \chi$ with $\bar{u}$ constant, then $q\,\mathrm{TV}_h(t) = qc\, e^{\lambda t}\, \mathrm{TV}_h(\chi)$ and $\max_e q|\Delta_e u| = qc\, e^{\lambda t} \max_e |\Delta_e \chi|$. The display is faithful, $D_h = 0$, exactly for $t \le t^* = \lambda^{-1} \log\bigl(1/(qc \max_e |\Delta_e \chi|)\bigr)$; after t^*, $\mathbb{E}_\theta P_h \le |E|$ while $D_h \ge q\,\mathrm{TV}_h - |E|$ grows exponentially.*

2. *(End state.) If $u(t) \to u_\infty$, then $\mathbb{E}_\theta P_h(t) \to \sum_e \varphi(q|\Delta_e u_\infty|)$ and $D_h(t) \to D_h(u_\infty)$. The display ends faithful if and only if $q \max_e |\Delta_e u_\infty| \le 1$. In particular it ends in one colour when u_∞ is constant, and it is aliased for all large t when $q \max_e |\Delta_e u_\infty| > 1$.*

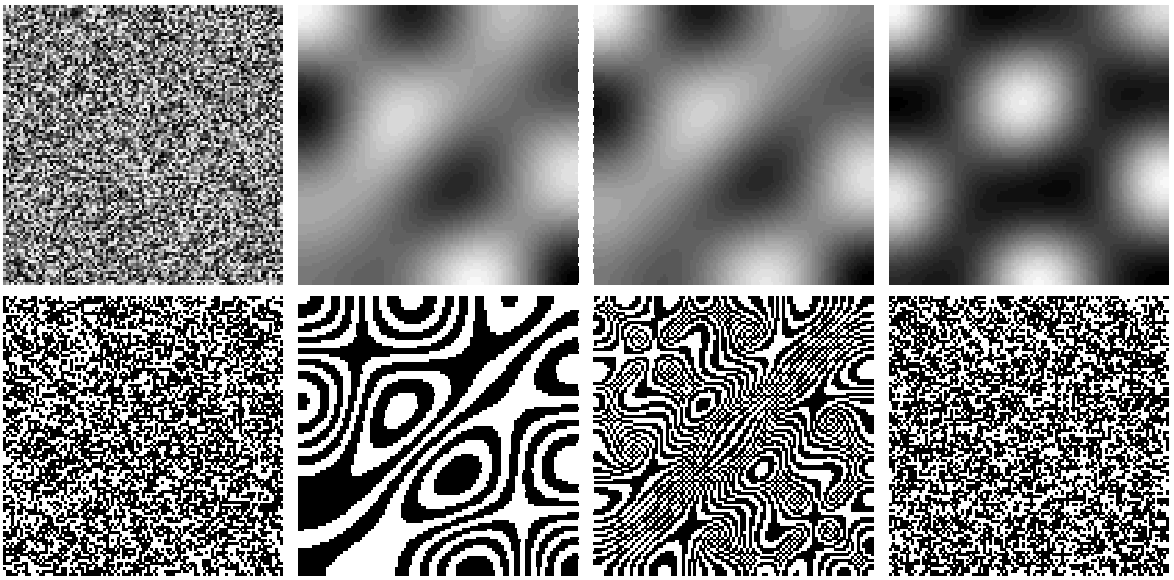
3. *(Flicker.) For any two fields u, u' on the M nodes, with θ uniform,*

$$\mathbb{E}_\theta \tfrac{1}{M} \#\{x : b_\theta[u](x) \ne b_\theta[u'](x)\} = \tfrac{1}{M} \textstyle\sum_x \varphi\bigl(q|u(x) - u'(x)|\bigr). \tag{6}$$

   If $u(t) \to u_\infty$ then, for every θ with $qu_\infty(x) + \theta \notin \mathbb{Z}$ at all nodes, the picture $b_\theta[u(t)]$ is eventually constant.

Proof. (1) TV_h and the jumps do not see the constant $\bar{u}$ and are homogeneous of degree one in $c\, e^{\lambda t}$; the rest is Corollary 3.2. (2) φ and ψ are continuous and the sums are finite; the criterion is Corollary 3.2 applied to u_∞. A constant field has a uniform picture. (3) The proof of Theorem 3.1(2) applies verbatim to the pair $(u(x), u'(x))$ in place of the two ends of an edge: a time step is an edge too. Finally $\lfloor \cdot \rfloor$ is locally constant off $\mathbb{Z}$. □

Part (2) is the difference with the heat equation. Neumann heat flow tends to a constant, so every jump eventually falls below one quantum and the frames end in a single colour. A Turing pattern tends to a non-constant steady state, and if its jumps exceed one quantum the aliasing never ends.



(a) Noise, $t = 0$ (b) Smooth, $t = 200$ (c) Breakdown, $t = 350$ (d) Static, $t = 1000$

Figure 4: The glycolysis model (5) with the parameters of [13, Fig. 4]. Top: u in grey, rescaled to its own range in each frame. Bottom: the parity of $\lfloor 10^4 u \rfloor$. The field is noise only at $t = 0$; after that it is a smooth pattern that grows and settles into spots. The parity picture is faithful only in (b). In (c) it shows moiré, and in (d) it is static, although the field is a few smooth spots.

Table 3 and Figure 4 show four stages. *Noise* ($t = 0$): the initial perturbation jumps by up to 99 quanta between neighbours, so the first frame is aliased. Diffusion removes these short modes within ten time units. *Smooth* ($10 \lesssim t < t^*$): every jump is below one quantum, $D_h = 0$, and the picture is the exact contour map, with $\mathbb{E}_\theta P_h = q\,\mathrm{TV}_h$. By $t \approx 100$ the slowest stable modes have decayed and the unstable mode dominates. It keeps its shape and grows, so the picture looks stable while its rings multiply (5 bands at $t = 100$, 23 at $t = 250$). A least-squares fit of $\log q\,\mathrm{TV}_h$ on $[200, 350]$ gives the rate 0.01094, against $\lambda = 0.01103$ from linear theory. Counting contours measures the leading Turing eigenvalue. *Breakdown*: the largest jump crosses one quantum at $t^* = 257.8$, as part (1) predicts from the fitted rate and the jump at $t = 250$. The deficit D_h opens and the zone-plate moiré of Section 5 appears around the extrema. *Static*: the amplitude saturates at a range of about 0.56 (the colour scale of [13, Fig. 4c]), and the largest jump of the steady state is 0.0270, that is 270 quanta. $\mathbb{E}_\theta P_h$ is pinned near $|E|/2$, the block entropy is at its maximum, and D_h carries 99.5% of $q\,\mathrm{TV}_h$. By part (2) the static is permanent, and for this grid it is permanent for every $q > 1/0.0270 \approx 37$. A faithful contour map of the final spots on this grid has at most about 20 bands. By part (3), the static still changes from frame to frame long after the pattern has formed: between $t = 1500$ and 2000 the field moves by at most $1.8 \cdot 10^{-4}$, under two quanta, and 43% of the cells change colour. Only after $t \approx 2500$, when the field moves by less than 10^{-6}, does the static freeze.

The two examples are mirror images. In the heat equation the field loses structure monotonically, and aliasing *manufactures* a rise of the complexity proxies. In the Turing instability the field gains structure: symmetry breaks, and $q\,\mathrm{TV}_h$ grows by a factor of about

t	range of u	bands	$q\,\mathrm{TV}_h$	$\sum_e \varphi(q\lvert\Delta_e u\rvert)$	D_h	$\max_e q\lvert\Delta_e u\rvert$	flicker
0	$1.00 \cdot 10^{-2}$	99	677626	10120	667506	98.96	—
10	$8.00 \cdot 10^{-4}$	8	1724	1724	0	0.36	0.5012
50	$4.77 \cdot 10^{-4}$	4	1159	1159	0	0.22	0.4908
100	$5.40 \cdot 10^{-4}$	5	1299	1299	0	0.24	0.5106
150	$7.74 \cdot 10^{-4}$	8	1936	1936	0	0.35	0.5219
200	$1.31 \cdot 10^{-3}$	13	3217	3217	0	0.56	0.4984
250	$2.32 \cdot 10^{-3}$	23	5519	5519	0	0.92	0.5351
300	$4.05 \cdot 10^{-3}$	41	9546	8929	617	1.56	0.4918
350	$7.05 \cdot 10^{-3}$	71	16551	10068	6483	2.70	0.4982
400	$1.22 \cdot 10^{-2}$	123	28716	10589	18126	4.68	0.5004
500	$3.69 \cdot 10^{-2}$	368	86416	9899	76517	14.13	0.4952
600	$1.10 \cdot 10^{-1}$	1097	258238	10104	248134	42.90	0.4977
800	$5.31 \cdot 10^{-1}$	5309	1399588	10138	1389450	228.75	0.4984
1000	$5.92 \cdot 10^{-1}$	5915	1853006	10144	1842862	268.60	0.4973
1500	$5.61 \cdot 10^{-1}$	5606	1967611	10173	1957438	269.68	0.5002
2000	$5.61 \cdot 10^{-1}$	5606	1967612	10174	1957438	269.66	0.4254
2500	$5.61 \cdot 10^{-1}$	5606	1967612	10174	1957438	269.66	0.0008
3000	$5.61 \cdot 10^{-1}$	5606	1967612	10174	1957438	269.66	0.0000

Table 3: The measures of Theorem 3.1 and Corollary 3.2 for Figure 4, $q = 10^4$, $|E| = 20200$. The Monte-Carlo average of P_h over 64 phases agrees with the formula column to 0.5% throughout. Flicker is the right side of (6) between consecutive rows; the fraction of cells that changed colour, averaged over 64 phases, agrees with it to 0.003.

1700 from $t = 50$ to the steady state. Yet after t^* the proxies are flat at their maxima, and aliasing *hides* the growth. They read the same for the growing pattern at $t = 600$, whose $q\,\mathrm{TV}_h$ is still 7.6 times below its final value, and for the frozen one at $t = 3000$. In both cases the deficit D_h of Corollary 3.2 is the exact measure of the part of the picture that is sampling.

7 What is not claimed, and what the lesson is

The number of components of the pattern, and the number of local maxima of u, are not monotone in general in two dimensions [3]; nor is the entropy of the value histogram of u. There are no eddies: for $t > 0$ the field is analytic with smooth nested isotherms. The lesson is methodological. Perimeter, block entropy and component counts of a thresholded simulation measure the sampling as much as the dynamics; the phase-averaging identity (3) is a diagnostic: by Corollary 3.2 the deficit $q\,\mathrm{TV}_h - \mathbb{E}_\theta P_h$ is exactly the part of the picture produced by the sampling, whether the field is decaying (Section 3) or growing (Section 6). Aliasing can manufacture apparent complexity and it can hide real complexity, and a thresholded picture alone cannot tell which. The practical rule is to choose the quantum against the largest jump the field will reach, not against the first frames: $q \max_e |\Delta_e u| \le 1$ at all times, or a quantum rescaled to the current range of the field, as a colour bar is. Real complexity requires the nonlinearity inside the dynamics: putting the modulus into the update rule instead of the display gives Wolfram's continuous cellular automata [2, ch. 4], coupled map lattices in the sense of [8], which do have localized structures; their interaction with reflecting walls is the subject of a companion paper [16].

A remark on functional information. Wong et al. [10] have proposed a law of increasing functional information: the functional information of a system increases when many configurations of the system undergo selection for one or more functions. Functional information, in

the sense of Szostak and Hazen [11, 12], is $I(E_x) = -\log_2 F(E_x)$, where $F(E_x)$ is the fraction of all configurations that perform a given function to degree at least E_x; it is defined only relative to a function, and the proposed law makes selection, not mere dynamics, the source of its growth. Quantized diffusion is a clean example of why the restriction to a function is necessary. The quantized heat frames have no function and undergo no selection; every quantity intrinsic to the field, the band count, the total variation, the gradient, decreases monotonically (Theorems 2.1 and 3.1); and yet the function-free complexity proxies that are commonly read as evidence of "ratcheting" complexity, perimeter, block entropy, component count, rise from zero to near their maxima before they fall. The rise is produced entirely by the observer's choices of quantum and grid, and Theorem 4.1 shows that these choices can make the block entropy anything between zero and its maximum. The random-phase identity (3) is the diagnostic that separates the two cases: a phase-averaged perimeter above $q\,\mathrm{TV}_h$ is sampling, not structure. The Turing example of Section 6 shows the converse failure. There structure does increase, through a symmetry-breaking instability with no selection involved, and the same proxies stop registering it at t^*. They report identical values for a pattern that is still growing and for one that is frozen. A system has to do more than produce pictures that look complex before the question of selection can even be asked; in the companion paper [16] we take automata that do sustain persistent localized structures and ask whether the walls of a box act on them as a selecting environment.

Declarations

Code availability. All scripts, the figures and the source of this article are public at `https://github.com/MiguelPerezI/heat-parity-ca`. The script `analyze.py` reproduces the frames and the measures of Table 2, `restest.py` the resolution test, `phase_theorem.py` checks Theorem 3.1 and produces Table 1, and `glyco.py` runs the glycolysis model of Section 6 and produces Table 3 and Figure 4. They need only Python and NumPy.

Funding. No funding was received for this work.

Competing interests. The author has no competing interests to declare.

Use of AI tools. A large language model (Claude, Anthropic) was used as an assistant in writing the analysis scripts, running the numerical checks and drafting the text. The author checked every statement, proof and number in the article and is responsible for its content.

About the author

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